FRANCISCO LOURENÇO

3D Data Specialist | ML & CV Engineer

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https://orcid.org/0000-0001-7081-5641 | Date of Birth: 23-04-1996

SUMMARY

Machine Learning and Computer Vision Engineer specialized in real-world visual data, 3D data, and end-to-end CV/ML systems. Experienced in designing, building, and deploying production-grade solutions across aerial imagery, street-level data, LiDAR, RGB-D, and industrial vision applications. Strong background in segmentation, geometry extraction, measurement systems, sensor fusion, 3D reconstruction, and dataset creation, including annotation workflow design and team coordination. Skilled at taking research-driven projects from data collection and prototyping to robust implementation and deployment in fast-paced environments.

WORK EXPERIENCE

3D Data Specialist | ML & CV Engineer
Freelance
Apr 2025 – Present | Coimbra, Portugal (Remote)

- Delivered end-to-end ML/CV solutions for aerial imagery, LiDAR, street-view, and geospatial applications, from data collection and annotation to model training, geometry extraction, measurement, and deployment.
- Built 3D and sensor-fusion pipelines for real-world spatial data, including LiDAR-camera colorization, georeferencing, orthophoto stitching, and reconstruction workflows.
- Designed dataset and annotation systems, including taxonomy definition, QA processes, and leading annotation teams for segmentation and estimation tasks.

Machine Learning Engineer
Anything World
Jan 2023 – Apr 2025 | London, UK (Remote)

- Designed and deployed ML pipelines for 3D shape classification, canonical alignment, and part segmentation used in the Animate Anything tool and Anything World API.
- Achieved 95.6% F1-score on the rotated ModelNet40 benchmark through a customized multi-view GCN architecture optimized for rotation invariance and computational efficiency.
- Developed a proprietary 3D rotation estimation system (Symmetry-Aware and Base-Plane-Aided) achieving 2.47° median angular error across 220+ characters.
- Built an optimized 3D shape segmentation pipeline with a GCN model reaching 88.8% mIoU on ShapeNet Part; reduced training and inference time significantly compared to the prior system.

3D Computer Vision Engineer
coatingAI
Sep 2021 – Jan 2023 | Barcelona, Spain

- Solely designed and developed a multi-camera RGB-D 3D scanning system for automated part inspection in powder coating lines.
- Built and deployed a full pipeline for 3D reconstruction, camera pose estimation, and point cloud fusion from stereo and depth data.

Junior Researcher
Institute of Systems and Robotics
Jan 2021 – Jun 2021 | Coimbra, Portugal

- Developed an anomaly detection system for glass bottle production lines, combining classical computer vision and machine learning methods.
- Worked with real-world industrial datasets under high-throughput and high-accuracy inspection constraints.

EDUCATION

MSc in Electrical and Computer Engineering
University of Coimbra
2015 – 2021 | Portugal

- Specialized in Computer Vision and Machine Learning.
- Thesis: *6DoF Object Pose Estimation from RGB-D Images using ML* — graded 19/20.

MSc Exchange Program
Aalto University
2018 – 2019 | Espoo, Finland

- Coursework focused on AI and Robotics.

PUBLICATIONS

Intel RealSense SR305, D415 and L515: Experimental Evaluation and Comparison of Depth Estimation
VISIGRAPP 2021

Evaluation of the Accuracy of Pose Estimation Based on Relative Pose
RECPAD 2021

TECHNICAL SKILLS

Python · PyTorch · TensorFlow · OpenCV · Open3D · C++ · SQL · HTML · CSS · MatLab · ROS · Blender · CAD · Docker

FIELDS OF INTEREST

Machine Learning · Computer Vision · Point Clouds · Data Science · Mathematics · Coding · Health Care · Robotics · Automation and Control · Self Driving Vehicles · 3D Animation